

MEI DA65646 ASIC — Software-Visible Map

Panasonic CF-VEW211 PCMCIA Sound Card · recon of the Matsushita custom ASIC (IC6, 100-pin QFP) · probed empirically with VEWGO project tooling (probes/VEWASIC.C) on an IBM PC110, 2026-07-09. All reset values are true power-on states (clean socket power-up). vew21xgo project — zikolas.

1 · Address decode summary

Space	Decode	Contents
Attribute 0x000–0x1FE (even)	256 bytes	CIS shadow RAM — host-writable, volatile; loaded from the 93LC56 EEPROM at power-up (load fails on this unit → constant 0x07 fill)
Attribute 0x200–0x208 (even)	5 registers	Config register file (§3)
Attribute 0x20A and above	none	Reads 0xFF (not registers)
Attribute space, globally	mirrors every 0x1000	Only A0–A11 are decoded
Common memory (0–256 KB scanned)	nothing	Reads 0xFF, no write latch — the on-board 32K SRAM (CXK58257) is not host-visible
I/O base+0 ... base+3	reads 0x00, no bits latch	ASIC ports — write-only or unimplemented; function unknown
I/O base+4 ... base+7	CS4231A codec	IAR / IDR / Status / PIO data
I/O base+8, base+9	no decode (0xFF)	Despite the CIS declaring a 10-byte I/O range
I/O 0x388–0x38B	YMF262 (OPL3)	Mapped in every configuration index

2 · CIS shadow

- Loaded from the on-board **93LC56** serial EEPROM at socket power-up. On this unit the load fails and the shadow fills with a constant **0x07**.
- Host writes to even attribute addresses 0x000–0x1FE latch immediately; contents survive warm reboots (the card keeps power) but not socket power-down. Nothing above 0x1FE accepts writes except the config registers.
- Writes do **not** reach the EEPROM (verified by power cycling). No write-through/unlock found yet — the undeclared vendor bits (§3) are the remaining candidates.
- Accurite BURNER.EXE (HeadstartCard/Z86M17 protocol) cannot reach the EEPROM either — verified negative: its reads see open bus (0xFF) and its erase/write verify against that.

3 · Config register file (attribute space, even addresses)

Power-on state: C0 00 00 00 00, then 0xFF from 0x20A.

Addr	Name	Reset	Bits that latch	Notes
0x200	COR	0xC0	0xEF (bit4 never latches)	Powers up in SRESET (bit7) + LevIREQ (bit6). Bit6 reads back 1 always — level-mode IRQ is hardwired. Config index in bits 5–0.
0x202	CCSR	0x00	0x08 only	Only the Audio bit exists (gates mixer audio onto #SPKR — confirmed audibly). IntrAck / Intr / PwrDwn unimplemented.
0x204	vendor	0x00	0x01 (1 bit)	Undeclared in the CIS. Function unknown.
0x206	vendor	0x00	0xFE (7 bits)	Undeclared. Seven writable bits — prime candidate for an FM attenuator / volume or control latch.
0x208	vendor	0x00	0x0F (4 bits)	Undeclared. Function unknown.

Note: the 0x20A+ region reads a stuck 0xFF; a naive bit-set test false-positives there — they are not registers.

4 · COR index decode

Index	Codec decodes?	COR readback	Note
0x00 / 0x01 / 0x10	no	0x40 / 0x41 / 0x40	unconfigured / undefined
0x20	yes @ 0x530	0x60	default entry
0x21	yes @ 0xE80	0x61	

Index	Codec decodes?	COR readback	Note
0x22	yes @ 0xF40	0x62	
0x23	yes @ 0x604	0x63	
0x24	yes @ 0x530	0x64	index bit2 is ignored — decode is partial
0x2F / 0x3F	not at 0x530	0x6F (bit4 dropped)	consistent with COR bit4 not implemented

Effective decode $\approx$ index & 0x23: bit5 = configured, bits 1–0 select the codec base. The CIS config table is validated in silicon.

5 · Confirmed audio architecture

- **PCM:** host → CS4231A PIO → codec DAC → mixer → output amp. Codec volume (I6/I7) works over the full range. PIO protocol gotchas: PRDY is unusable for pacing on this card (reads ready while overfeed is dropped) — pace by PIT; and each sample only commits to the FIFO when the Status register is read after writing it.
- **FM:** YMF262 → YAC512 DAC → analog, summed **after** the codec into the output amp. Unaffected by every codec mixer control (proven by muting all inputs including MODE2 LINE and MONO). No volume control found yet — the vendor bits (\$3) and the write-only ports base+0..+3 are the remaining candidates; the board carries a TC4W66F analog switch in the output section that something must drive.
- **#SPKR:** CCSR bit3 drives a 1-bit version of the card audio onto the socket speaker pin (host side: PCIC Misc Control 1 bit4 routes it to the laptop speaker). Fixed level, harsh by nature.

6 · Open questions

- Function of vendor bits 0x204.0, 0x206.7–1, 0x208.3–0 — FM volume/mute? amp gate? EEPROM write-enable? (Needs a listening pass with the FM test note, and a write-persistence pass.)
- Function of the write-only(?) I/O ports base+0 ... base+3.
- Why the CIS declares 10 I/O bytes while the card decodes 8.
- Which register (if any) drives the TC4W66F analog switch.

Appendix · Raw probe output (VEWASIC.C, 2026-07-09)

VEWASIC - CF-VEW211 MEI ASIC (DA65646) software-visible map

=====

power-up: IF status=7F (READY=1)

[1] attribute space map (256-byte blocks, even addresses):

```
attr 00000: const fill 07
attr 00200: DATA (sum=FAC5, first=C0 00 00 00)
attr 00400: const fill FF
attr 01000: const fill 07
attr 01200: DATA (sum=FAC5, first=C0 00 00 00)
attr 01400: const fill FF
attr 02000: const fill 07
attr 02200: DATA (sum=FAC5, first=C0 00 00 00)
attr 02400: const fill FF
attr 03000: const fill 07
attr 03200: DATA (sum=FAC5, first=C0 00 00 00)
attr 03400: const fill FF
attr 04000: const fill 07
attr 04200: DATA (sum=FAC5, first=C0 00 00 00)
attr 04400: const fill FF
attr 05000: const fill 07
attr 05200: DATA (sum=FAC5, first=C0 00 00 00)
attr 05400: const fill FF
attr 06000: const fill 07
attr 06200: DATA (sum=FAC5, first=C0 00 00 00)
attr 06400: const fill FF
attr 07000: const fill 07
attr 07200: DATA (sum=FAC5, first=C0 00 00 00)
attr 07400: const fill FF
attr 08000: const fill 07
attr 08200: DATA (sum=FAC5, first=C0 00 00 00)
attr 08400: const fill FF
attr 09000: const fill 07
attr 09200: DATA (sum=FAC5, first=C0 00 00 00)
attr 09400: const fill FF
attr 0A000: const fill 07
attr 0A200: DATA (sum=FAC5, first=C0 00 00 00)
attr 0A400: const fill FF
attr 0B000: const fill 07
attr 0B200: DATA (sum=FAC5, first=C0 00 00 00)
attr 0B400: const fill FF
attr 0C000: const fill 07
attr 0C200: DATA (sum=FAC5, first=C0 00 00 00)
attr 0C400: const fill FF
attr 0D000: const fill 07
attr 0D200: DATA (sum=FAC5, first=C0 00 00 00)
attr 0D400: const fill FF
attr 0E000: const fill 07
attr 0E200: DATA (sum=FAC5, first=C0 00 00 00)
attr 0E400: const fill FF
attr 0F000: const fill 07
attr 0F200: DATA (sum=FAC5, first=C0 00 00 00)
attr 0F400: const fill FF
```

[2] attribute write-latch extent (even addresses):

latches: 000:Y 040:Y 080:Y 0C0:Y 100:Y 140:Y 180:Y 1C0:Y 1FE:Y 210:n 220:n 240:n 280:n 300:n 3FE:n 400:n 500:n 7FE:n

[3] config register file 0x200-0x21E (even bytes, reset state):

```
C0 00 00 00 00 FF FF FF FF FF FF FF FF FF FF
COR @200: reset=C0 bits that latch=EF codec after=DEAD
CCSR @202: reset=00 bits that latch=08 codec after=DEAD
+204 @204: reset=00 bits that latch=01 codec after=DEAD
+206 @206: reset=00 bits that latch=FE codec after=DEAD
+208 @208: reset=00 bits that latch=0F codec after=DEAD
+20A @20A: reset=FF bits that latch=FF codec after=DEAD
+20C @20C: reset=FF bits that latch=FF codec after=DEAD
+20E @20E: reset=FF bits that latch=FF codec after=DEAD
+210 @210: reset=FF bits that latch=FF codec after=DEAD
```

[4] I/O ports at idx 0x20 (base 0x530):

```
read: 530=00 531=00 532=00 533=00 [codec 534-537: 80 80 80 80] 538=FF 539=FF
port 530: reset=00 bits-latch=00 codec=ok
port 531: reset=00 bits-latch=00 codec=ok
port 532: reset=00 bits-latch=00 codec=ok
port 533: reset=00 bits-latch=00 codec=ok
port 538: reset=FF bits-latch=FF codec=ok
port 539: reset=FF bits-latch=FF codec=ok
```

[5] COR index walk (codec IAR at expected base; FF = no decode):

```
idx 00 -> base 530: +4=FF +6=FF COR readback=40
idx 01 -> base 530: +4=FF +6=FF COR readback=41
idx 10 -> base 530: +4=FF +6=FF COR readback=40
idx 20 -> base 530: +4=80 +6=80 COR readback=60
```

```
idx 21 -> base E80: +4=80 +6=80 COR readback=61
idx 22 -> base F40: +4=80 +6=80 COR readback=62
idx 23 -> base 604: +4=80 +6=80 COR readback=63
idx 24 -> base 530: +4=80 +6=80 COR readback=64
idx 2F -> base 530: +4=FF +6=FF COR readback=6F
idx 3F -> base 530: +4=FF +6=FF COR readback=6F
```

[6] common memory scan (16KB blocks, first 256KB):

```
common 00000: const (first=FF) write-latch=no
common 04000: const (first=FF) write-latch=no
common 08000: const (first=FF) write-latch=no
common 0C000: const (first=FF) write-latch=no
common 10000: const (first=FF) write-latch=no
common 14000: const (first=FF) write-latch=no
common 18000: const (first=FF) write-latch=no
common 1C000: const (first=FF) write-latch=no
common 20000: const (first=FF) write-latch=no
common 24000: const (first=FF) write-latch=no
common 28000: const (first=FF) write-latch=no
common 2C000: const (first=FF) write-latch=no
common 30000: const (first=FF) write-latch=no
common 34000: const (first=FF) write-latch=no
common 38000: const (first=FF) write-latch=no
common 3C000: const (first=FF) write-latch=no
```

[7] I/O decode footprint 0x500-0x53F (window widened):

```
500: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
510: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
520: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
530: 00 00 00 00 40 00 CC 00 FF FF FF FF FF FF FF FF
```

done - run VEW21XG0 to re-heal the CIS + restore normal config.